

# Isuru Rajapakshe

Los Angeles, CA

+1 (213) 643-9972 | rmisur@gmail.com | LinkedIn | GitHub | Portfolio

## Summary

Electrical Engineer specializing in sensing systems, signal processing, and embedded development. Experience spanning LiDAR perception systems, neuroscience research, experimental hardware validation, and real-time data processing. Strong background in Python-based analysis, machine learning, and sensor characterization for complex engineering systems.

## Technical Skills

**Programming:** Python, C, JavaScript, Shell Scripting

**Signal Processing & Analysis:** FFTs, Spectral Analysis, Filtering, Sensor Characterization, EEG Analysis

**Embedded Systems:** Zephyr RTOS, Arduino, nRF52, Microcontroller Interfacing

**Machine Learning:** PyTorch, MATLAB

**Data & Visualization:** NumPy, Pandas, Plotly, Open3D

**Engineering Tools:** Git, SolidWorks, OrCAD, Proteus, Firebase

## Experience

### Development Engineer

May 2025 – Present

*SiLC Technologies*

*Los Angeles, CA*

- Develop and validate FMCW LiDAR systems for perception and ranging applications.
- Build Python pipelines for point cloud processing, range precision analysis, and sensor characterization.
- Conduct testing and validation across optical, electrical, and embedded subsystems.
- Develop tools for automated visualization and performance analysis.

### Course Instructor / Instructional Assistant

Jun 2024 – Dec 2025

*California State University Los Angeles*

*Los Angeles, CA*

- Mentored undergraduate engineering students in digital circuits, and biomedical devices
- Guided interdisciplinary teams through hardware-software integration and project delivery.

### Software Engineer

Jul 2021 – Jul 2022

*Insync Information Technologies*

*Colombo, Sri Lanka*

- Led development of a network automation platform using the MERN stack.
- Designed software workflows to automate network configuration, monitoring, and management tasks.

### Electronic Engineering Intern

Jun 2019 – Dec 2019

*Lanka Electronics*

*Colombo, Sri Lanka*

- Developed trajectory planning and welding control features for a 6-DOF robotic arm in C#.
- Improved robotic positioning accuracy to approximately 3 mm.

## Projects

### EEG Biomarker Analysis for rTMS Therapy

Dec 2023 – May 2025

*Research Assistant, California State University, Los Angeles*

- Developed MATLAB and Python pipelines for EEG preprocessing, spectral analysis, and feature extraction.
- Investigated EEG biomarkers associated with rTMS treatment response in Major Depressive Disorder.
- Applied signal processing and statistical analysis to multi-channel neural recordings.

### Gesture-Controlled Music Generation System

Jun 2024 – Aug 2024

*Student Mentor, BOOST Program, California State University, Los Angeles*

- Led a five-member team in developing a gesture-controlled music generation device using real-time hand tracking.
- Integrated embedded hardware, software, and user interaction design into a functional prototype.

### LipRead — Visual Speech Recognition Pipeline

Nov 2023 – Dec 2023

*Graduate Course Project, California State University, Los Angeles*

- Developed a PyTorch-based visual speech recognition pipeline using GRUs.
- Achieved 66% word prediction accuracy on the evaluation dataset.

### Non-Intrusive Load Monitoring (NILM) System

Feb 2020 – Jul 2021

*Undergraduate Capstone Project, University of Moratuwa*

- Developed an appliance-level energy monitoring system using embedded sensing and machine learning.
- Designed power metering circuitry in OrCAD and developed Firebase-based monitoring tools.
- Achieved 90% kettle and 95.5% microwave classification accuracy.

## Education

**Master of Science in Electrical Engineering** — GPA: 4.0/4.0

Aug 2023 – May 2025

*California State University, Los Angeles*

*Los Angeles, CA*

**Bachelor of Science in Engineering (Hons.)** — GPA: 3.57/4.2

Oct 2016 – Jun 2021

*University of Moratuwa*

*Sri Lanka*